Changelog

All notable project changes are documented here.

The project does not currently use tagged releases, so this changelog follows the development history on the `main` branch.

[Unreleased]

Known limitations

- Current traffic reservations are atomic across the complete navigation leg and therefore reduce concurrency.
- In 4-robot workloads, only 1–2 tasks may be actively progressing at some times because large parts of the graph are reserved by one robot.
- Head-on collisions are reduced but not yet eliminated in all dynamic cases.
- Resource occupancy is not yet direction-aware.

2026-11-04 — Fleet traffic coordination and balanced workload generation

Commit: `51b827b`

Added

- Central `traffic_manager` node launched with the fleet.
- `TrafficNavigate.action` as the coordination interface between robot agents and the traffic manager.
- Warehouse traffic graph generated from the same layout source as the Gazebo world and maps.
- Explicit graph nodes for stations, asile intersection and corridors.
- Capacity-one resources for aisles, corridors and approach areas.
- Resource-conflict definitions between crossing rack aisles.
- A* graph routing that avoids resources currently reserved by another robot.
- Atomic per-leg resource reservation.
- Dense route waypoints passed to Nav2 `NavigateThroughPoses`.
- Balanced task-generation mode using `--balanced`.
- Argument validation for task count, interval, pickup, and drop names.

Changed

- Robot agents now request coordinated navigation from `/fleet/traffic_navigate` instead of sending each mission leg directly to Nav2.
- `generate_world.py` now writes `traffic.yaml` in addition to the world, station configuration, and occupancy maps.
- The navigation map now keeps the real gaps between rack rows traversable; rack traffic restrictions are handled by the traffic manager rather than by blocking the full rack region in the occupancy map.

- Task endpoints are explicit graph nodes, reducing unreserved diagonal shortcuts between free-space poses and lane nodes.
- Traffic-graph waypoint spacing reduced to 0.30 m.
- Random-task utility can produce reproducible balanced pick/drop coverage.

Current behavior / trade-off

- Head-on conflicts are substantially reduced by capacity-one lane reservations.
- The implementation intentionally holds the complete set of constrained resources needed by one navigation leg until that leg completes.
- This prevents many resource deadlocks by construction but lowers fleet throughput and can leave only one or two robots actively progressing.
- Some dynamic head-on/conflict cases are still possible and remain an active development topic.

2026-11-03 — Battery-aware allocation, dock management, and conservative rack routing

Commit: `01e73c8`

Added

- Battery awareness in task assignment.
- Mission energy estimation including return to a dock.
- Nearest-free-dock selection and dock reservation.
- Dock ID reporting in fleet/robot monitoring.

Changed

- Navigation map temporarily blocked the complete rack region so robots stayed on wider outer lanes.

Superseded

- The whole-rack blocking strategy was replaced on 2026-11-04 by the generated lane graph and traffic-manager reservations, allowing rack aisles to participate in routing.

2026-11-03 — Fleet manager, robot agents, and mission FSM

Commit: `8d8adff`

Added

- Per-robot `robot_agent` nodes.
- Custom ROS interfaces for robot state, fleet state, task submission, docking, and mission execution.
- Fleet task queue and robot allocation.
- Pick/drop/charge mission state machine.
- Battery consumption and charging behavior.
- Automatic dock return.
- Fleet status monitoring.

- Random task generation.

2026-11-01 — Nav2 and AMCL navigation

Commit: `1e86266`

Added

- AMCL localization.
- Per-robot Nav2 navigation stacks.
- Goal-sending utility for navigation testing.
- Initial controller/path tuning.

Identified

- Independent robot planning could produce fleet-level collisions, motivating later traffic coordination.

2026-11-01 — Generated warehouse world and maps

Commit: `57924a4`

Added

- Python generator for the Gazebo warehouse world.
- Generated station/layout configuration.
- Generated occupancy-map assets for navigation.

2026-10-31 — Multi-robot simulation foundation

Commit: `6fe21e0`

Added

- Warehouse SDF environment.
- AMR URDF/Xacro model.
- Parameterized multi-robot spawning.
- Gazebo and RViz visualization.
- Initial TF verification for multiple robots.

2026-10-31 — Initial repository

Commit: `f045be7`

Added

- Initial project structure and test commit.